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Clinical characteristics and outcomes of patients with Guillain–Barré and acquired CNS demyelinating overlap syndrome: a cohort study based on a literature review

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Background: Some patients with Guillain–Barré syndrome (GBS) also have acquired demyelination of the central nervous system (CNS) (i.e. acquired demyelinating syndrome, ADS). Often, the overlap of GBS and ADS is overlooked. Therefore, we evaluated case reports of GBS/ADS overlap syndrome.

Methods: We mainly performed website-based research based on articles in cases presented with GBS/ADS overlap syndrome. A total of 66 cases were included. Clinical and prognosis data were analyzed.

Results: A total of 85% of patients with simultaneous or consecutive occurrence of GBS and ADS were identified within 4 weeks of the initial diagnosis. Transverse myelitis (TM) (32%) was the most common ADS found in GBS/ADS. Patients with Miller Fisher syndrome (MFS)/ADS overlap syndrome had greater female predominance, mean age, frequency of onset at the same time period, or within a short period, and percentage of sole involvement of the subtentorial region. The outcome was favorable based on the functional status in 74% of patients. The sensory level (OR = 0.182, 95% CI = 0.055–0.598; $P = 0.005$) was the best predictor of a poor outcome, while visual deficit (OR = 4.667, 95% CI = 1.187–18.352; $P = 0.027$) predicted a favorable outcome.

Conclusion: The ADS in GBS are diverse, CNS demyelinating may occur at any time, but early in the GBS course (and vice versa). MFS/ADS overlap syndromes is more common. The prognosis is generally good, but patients with sensory level deficit are likely to have a poor prognosis. The features of MFS/other CIS may better reflect involvement of the brainstem in MFS itself, rather than ADS in autoimmune peripheral neuropathies.

Keywords: Guillain–Barré syndrome, Demyelinating diseases, Central nervous system, Overlap syndrome, Miller Fisher syndrome, Clinically isolated syndromes, Transverse myelitis

Introduction

Guillain–Barré syndrome (GBS) and acquired demyelination of the central nervous system (CNS) (i.e. acquired demyelinating syndrome, ADS) are classically viewed as distinct autoimmune diseases. Patients presenting with clinical, biochemical, and serological results of both diseases, either simultaneously or consecutively, have been repeatedly described and recognized in case reports.^{1–3} Overlap syndromes are defined as the association of two conditions in a single patient. However, the clinical spectrum of GBS/ADS overlap syndrome remains poorly established and is

often overlooked.⁴ Nonetheless, GBS/ADS overlap syndrome encompasses a wide spectrum of manifestations of both the CNS and peripheral nervous system (PNS). As the clinical spectrum of GBS/ADS overlap syndrome is heterogeneous, a tentative subdivision in distinct patient subgroups may predict a therapeutic outcome.³ Therefore, we performed a website search-based study to evaluate the incidence, clinical course, and outcomes of GBS/ADS overlap syndrome.

Methods

Search strategy

A literature search using the PubMed database (from January 1966 to October 2013), the Chinese biomedical research literature (from January 1999 to October 2013) and a web-based search engine (Google) was performed. Key search terms used were GBS, acute

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inflammatory demyelinating polyneuropathy (AIDP), acute motor axonal neuropathy (AMAN), acute motor sensory axonal neuropathy (AMSAN), Miller Fisher syndrome (MFS), optic neuritis (ON), clinically isolated syndromes (CIS), transverse myelitis (TM), neuromyelitis optica (NMO), multiple sclerosis (MS), and acute disseminated encephalomyelitis (ADEM). The search included all publication types, genders, and medical subsets; however, articles were restricted to be published in English.

All patients with a clinical and electrophysiological diagnosis of GBS/ADS overlap syndrome were included.⁵ Patients were first divided into six ADS groups according to the consensus definitions: (1) ON, (2) TM, (3) NMO, (4) MS, (5) ADEM,^{6,7} and (6) other forms of CIS. Clinically isolated syndromes was defined as a monofocal or multi-focal attack of CNS demyelination (in the optic nerve, brainstem, hemisphere, cerebellum, or spinal cord).⁶ Other forms of CIS were defined as an episode of demyelination not including ON or TM.^{6,7} Diagnoses were confirmed, and additional clinical details were extracted. All patients with the following features were excluded: (1) those who developed acute onset chronic inflammatory demyelinating polyneuropathy (A-CIDP); (2) those who were unable to provide the necessary clinical information; (3) those with evidence of cerebral vasculitis, granulomatous disease, or presumed infectious encephalitis; and (4) those with metabolic disorders, vasculitis, and in whom non-specific magnetic resonance imaging (MRI) abnormalities were not suggestive of demyelinating disease. Additional studies were performed by searching reference lists of retrieved articles and manually searching the main neurology journals. We contacted case authors to identify additional published or unpublished data if possible. We used Cohen's kappa statistic (κ) to assess agreement between the two reviewers (ZM and XH).

Data extraction

A total of 66 cases were found in original case reports published between 1976 and 2012^{1-3,8-53} (including one published case by ZM *et al.* in China; the treatment for the patient was carried out according to the protocol approved by Institutional Ethical Committee of Affiliated Nationalities Hospital, Guangxi Medical University). The agreement between reviewers for inclusion of articles was 93.8% ($\kappa = 0.89$; 95% CI, 0.82–0.97).⁵⁴ Detailed characteristics and medical information of selected patients are described below. Comprehensive clinical and laboratory information was reviewed by a group of neurologists. Cohen's kappa for inter-rater agreement was 0.77 (95% CI, 0.63–0.90; substantial agreement). The demographic, clinical, and characteristics

data were summarized by frequency (%), and continuous data were expressed as mean $\pm$ standard deviation (SD) or median. MRI of brain lesions was analyzed by distribution in the supratentorial or subtentorial brain regions, or both. We defined patients as GBS (non-Miller Fisher syndrome, non-MFS) or MFS when they fulfilled the diagnostic criteria.^{5,55} MFS is a very unusual variant of GBS (compared with other types of GBS) and has clinical and electrophysiological features that would not meet criteria of standard GBS (i.e. AIDP, AMAN, and AMSAN).

To facilitate comparison with each study, we standardized our outcome criteria according to a scale adapted from Hughes *et al.*⁵⁶ Due to inherent characteristics of case reports, in which results of common symptoms/signs/tests usually tend to be presented regardless of positive or negative signs, we calculated the frequency of clinical, biological, and neuroimaging data in three ways. First, the motor and sensory symptoms were the most common symptoms affected in GBS; therefore, cases with missing data for either sign were excluded from separated frequency analyses. (This exclusion will impact a change in the denominator.) Second, as most negative symptoms and/or signs are usually not reported in case reports, we coded the unreported symptoms and/or signs as 'negative.' Third, as case reports usually provide all the common results of biological and neuroimaging tests regardless of positive or negative results, cases with missing data regarding biological and neuroimaging measures on factors of interest seemed to be 'not performed' and were finally excluded from these specific analyses.

The clinical, biochemical, electrophysiological, and neuroimaging factors obtained during admission were analyzed as possible predictors of outcome. A favorable outcome was defined as functional independence and the ability to ambulate independently (score of 0–2 on the Hughes scale).⁵⁶ The association between each of these variables and outcomes was examined. To further explore the natural course of both diseases and to assess whether both diseases presented simultaneously or consecutively, the first 4 weeks after onset was determined as the acute phase.

Statistical analysis

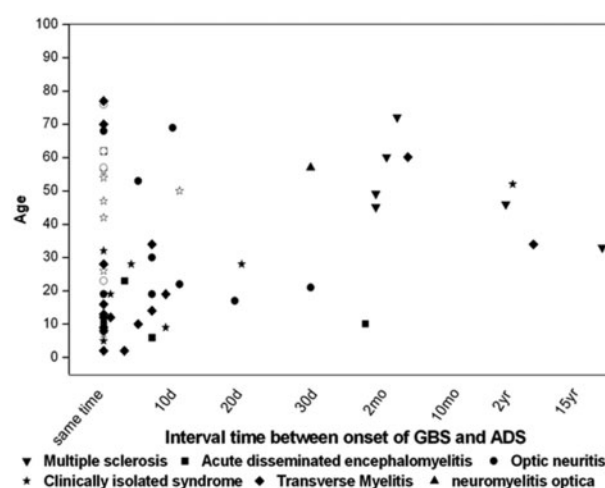
Statistical analysis was performed using SPSS statistical software, version 17.0 (SPSS, Inc., Chicago, IL, USA). Univariate factors predictive of outcomes (favorable or poor) were assessed using the non-conditional logistic regression method. All statistical tests were two-tailed. A *P* value less than 0.05 was considered statistically significant.

Results

Clinical characteristics

Demographic and clinical characteristics are presented in Table 1. Sixty-six (33 women and 33 men) patients were analyzed in this study. The mean age was 30.99 years old (range from 2 to 77 years old). We found that the occurrence of overlap syndrome after infection or vaccinations accounted for 47 cases (70%). In 57 cases (85%), GBS (including both non-MFS and MFS) and ADS occurred simultaneously or consecutively within 4 weeks after onset. In the rest nine cases (15%), the time interval between the occurrence of GBS and ADS was beyond 4 weeks after onset. Non-MFS/ADS overlap syndrome was identified in 55 patients (83%), while MFS/ADS overlap syndrome was identified in only 11 patients (17%). The most common subtypes of ADS identified were TM (32%), other forms of CIS (24%), and ON (20%). Twenty-nine patients (43%) initially manifested GBS/ADS overlap syndrome ‘at the same time’ (defined as within a very short period with no more than 1 day of the onset of GBS or ADS event), followed by 21 patients with GBS (31%) manifestations preceding ADS symptoms, whereas 17 patients (25%) presented with ADS symptoms preceding GBS. Figure 1 illustrates the trends in the time interval between the onset of GBS and the onset of each of the six subtypes of ADS (and vice versa). Miller Fisher syndrome overlapping with ADS (open shapes) showed a more frequent trend of co-occurrence than non-MFS groups with ADS (solid shapes).

Table 2 summarizes the clinical patterns among groups. Paresis (90%), sensory deficit (73%), cranial



Miller Fisher syndrome (MFS) (open shapes) occurring with ADS shows a more frequent trend ‘at the same time’, compared with non-MFS groups (solid shapes).

Figure 1 Trends in the time interval between the onset of Guillain-Barré syndrome (GBS) and the onset of six subtypes of acquired demyelinating syndrome (ADS) (and vice versa).

nerve palsies (55%), urinary disturbance (49%), and visual deficit (39%), together with a wide variety of other neurological signs, were the most frequent presentations. In patients with GBS/NMO, no cranial nerve palsy was observed; however, on average, it represented 55%; while extrapyramidal syndrome only presented in patients with GBS/MS but not in the other subtypes of the overlap syndrome.

Table 3 summarizes the biological and neuro-imaging data. Albuminocytologic dissociation was detected in 45 of 63 patients (71%). An IgG index greater than 0.68 was detected in one of five patients

Table 1 Main characteristics of patients with GBS/ADS overlap syndrome

	GBS/ON	GBS/TM	GBS/other CIS	GBS/NMO	GBS/MS	GBS/ADEM	Total
N (%)	13 (20%)	21 (32%)	16 (24%)	2 (3%)	7 (11%)	7 (11%)	66
Sex (M:F)	8:5	11:10	5:11	2:0	3:4	4:3	33:33
Mean age (range of years)	41.23 (17–76)	21.7 (2–77)	31.4 (5–56)	38.5 (20–57)	47.29 (26–72)	18.86 (6–62)	30.82 (2–77)
Infections or vaccinations	9 (69%)	16 (76%)	12 (75%)	2 (100%)	1 (14%)	7 (100%)	47 (70%)
Type of GBS (non-MFS:MFS)	9:4	21:0	9:7	2:0	7:0	7:0	55:11
Diagnosis series							
Primary GBS	4 (31%)	3 (14%)	6 (38%)	2 (100%)	1 (14%)	1 (14%)	17 (25%)
Primary central nervous system demyelination	3 (23%)	7 (33%)	2 (13%)	0	6 (86%)	3 (43%)	21 (31%)
Simultaneous initial diagnosis	6 (46%)	11 (52%)	8 (50%)	0	0	3 (43%)	29 (43%)
Interval between primary to secondary diagnosis							
≤4 weeks	13 (100%)	19 (91%)	15 (94%)	2 (100%)	2 (29%)	6 (86%)	57 (86%)
>4 weeks	0	2 (10%)	1 (6%)	0	5 (71%)	1 (14%)	9 (14%)
Immune therapy	10 (77%)	13/13 (100%)	12/15 (80%)	0/1	4/5 (80%)	7 (100%)	46/54 (85%)
Mean follow-up (SD), months	5.5 (4.2)	11.7 (6.6)	7.5 (5.8)	60 (0)	5.4 (4.3)	7.2 (9.5)	9.4 (10.6)
Outcomes							
Favorable outcome	12 (92%)	11 (55%) 1 drop)	14 (93%) 1 drop)	1 (50%)	3 (43%)	6 (86%)	47/64 (73.4%)
Poor outcome	1 (8%)	9 (45%)	1 (7%)	1 (50%)	4 (57%)	1 (14%)	17/64 (27%)
Died	1 (8%)	0	0	0	2 (29%)	1 (14%)	4/64 (6%)

GBS: Guillain-Barré syndrome; ADS: acquired demyelinating syndromes; ON: optic neuritis; CIS: clinically isolated syndromes; TM: transverse myelitis; NMO: neuromyelitis optica; MS: multiple sclerosis; ADEM: acute disseminated encephalomyelitis; MFS: Miller Fisher syndrome.

Data are expressed as mean (SD), median (range), or percent.

(20%). Oligoclonal bands were found to be positive in seven of 24 patients (29%). Visual evoked potential (VEP) abnormality was identified in three of 10 patients (30%). Anti-ganglioside antibodies were detected in seven patients (7/18; 39%). In the acute phase, 12 (12/43; 28%) patients had focal or multifocal lesions detectable by MRI in the supratentorial region, 8 (8/43; 19%) in the subtentorial region, and 4 (4/43; 9%) in both the supra- and subtentorial region. No positive findings of brain MRI were found in patients with GBS/ON, GBS/TM, or GBS/NMO. Abnormality in the spinal cord MRI was detected in 22 of 31 patients (71%).

Prognosis analysis

Immunotherapies were prescribed in 85% of patients with ADS or GBS presentation, mostly prednisone and intravenous immunoglobulin. The outcome of GBS/ADS overlap syndrome was good in 48 of 65 patients (74%, Table 1) during follow-up (median 10.7 months) using the Hughes scale. Only four patients died (one with GBS/ON overlap syndrome, two with GBS/MS, and one with GBS/ADEM) in the deteriorating stage. Of seven patients who did not receive immunotherapy, six recovered well and only one with GBS/MS overlap syndrome had a poor outcome. No significant factors from the clinical, biochemical, electrophysiological, and neuroimaging results were identified (data not shown). Only the sensory level (OR = 0.182, 95% CI = 0.055–0.598; $P = 0.005$) was a predictor of poor outcome, while the visual deficit (OR = 4.667, 95% CI = 1.187–18.352; $P = 0.027$) predicted a favorable outcome.

Subgroup analysis

Since the course between groups of MFS/ADS overlap syndrome and groups of non-MFS/ADS

overlap syndrome seemed not to be homogeneous, we performed further analyses between these subgroups. As shown in Table 4, patients with MFS/ADS overlap syndrome had a greater female predominance than those with non-MFS/ADS overlap syndrome. In addition, the mean age of patients with MFS/ADS overlap syndrome was significantly older than those with non-MFS/ADS overlap syndrome ($P = 0.013$). The frequency of co-occurrence at initial onset was approximately 3-fold more than that in the non-MFS/ADS overlap syndrome ($P = 0.004$), and 86% (6/7; all six cases were MFS/other CIS) of patients with MFS/ADS overlap syndrome showed sole involvement of the subtentorial region when compared with only 3% in patients with non-MFS/ADA overlap syndrome ($P = 0.000$). Moreover, patients with MFS/ADS overlap syndrome presented with a much more favorable outcome than those with non-MFS/ADS overlap syndrome ($P = 0.014$). However, patients with MFS/ADS overlap syndrome had hardly no involvement of the supratentorial region and spinal cord ($P = 0.011$; $P = 0.007$).

Discussion

The present study is the first clinical cohort study regarding the spectrum of GBS/ADS overlap syndrome. Considering the limited number of cases reported in the literature, this study represents a relatively large cohort study. The results allowed us to characterize the different subgroups in GBS/ADS overlap syndrome by systematic clinical, neuroimaging, and laboratory approach, which reflects the complex interaction between PNS and CNS involvement. Predictor (sensory level) of poor outcome can be identified after a careful analysis. Our results indicate that patients with GBS/ADS overlap

Table 2 Clinical presentation in GBS/ADS overlap syndrome

	GBS/ ON = 13	GBS/ TM = 21	GBS/other CIS = 16	GBS/ NMO = 2	GBS/ MS = 7	GBS/ ADEM = 7	Total = 66
Sensory level	0	16 (76%)	2 (13%)	1 (50%)	3 (43%)	1 (14%)	23 (35%)
Sensory	10 (77%)	18 (86%)	10 (63%)	2 (100%)	7 (100%)	1 (14%)	48 (73%)
Motor							
Paralysis	0	1/13 (8%)	1 (6%)	0	0	0	2/58 (3%)
Paresis	12 (92%)	12/13 (92%)	12 (75%)	2 (100%)	7 (100%)	7 (100%)	52/58 (90%)
Neuropathic pain	2 (15%)	8 (38%)	5 (31%)	1 (50%)	6 (86%)	3 (43%)	25 (38%)
Visual deficit	13 (100%)	0	5 (31%)	2 (100%)	3 (43%)	3 (43%)	26 (39%)
Cranial nerve palsies	11 (85%)	5 (24%)	12 (75%)	0	5 (71%)	3 (43%)	36 (55%)
Urinary disturbance	2 (15%)	18 (86%)	3 (18%)	2 (100%)	4 (57%)	3 (43%)	32 (49%)
Ataxia	6 (46%)	0	10 (63)	0	4 (57%)	1 (14%)	21 (32%)
Consciousness impairment	1 (8%)	0	1 (6)	0	4 (57%)	7 (100%)	13 (20%)
Respiratory failure	3 (23%)	6 (29%)	3 (19)	0	4 (57%)	2 (29%)	18 (27%)
Autonomic dysfunction	3 (23%)	5 (24%)	3 (18)	0	0	3 (43%)	14 (21%)
Meningeal reaction	0	0	2 (12%)	0	0	0	2 (3%)
Abnormal mental status	0	0	0	0	0	1 (14%)	1 (2%)
Seizure	0	0	0	0	0	2 (29%)	2 (3%)
Extrapyramidal syndrome	0	0	0	0	2 (29%)	0	2 (3%)

GBS: Guillain-Barré syndrome; ADS: acquired demyelinating syndromes; ON: optic neuritis; CIS: clinically isolated syndromes; TM: transverse myelitis; NMO: neuromyelitis optica; MS: multiple sclerosis; ADEM: acute disseminated encephalomyelitis. Data are expressed as mean (SD), median (range), or percent.

syndrome had typical clinical features of GBS and ADS in many aspects, thus representing an apparent overlap of the two conditions.^{7,57–60}

First, it is worthwhile to note that we found several demographic features of GBS/ADS overlap syndrome such as a younger age at onset and a balanced sex ratio, which are similar to those reported in typical global ADS studies; whereas these features were different from typical GBS features.^{59,61} In the majority of typical GBS studies, it is believed that the incidence increases linearly with age and that men are approximately 1.5 times more likely to be affected than women.^{59,62} The predominance of cases found at younger ages in our study could be attributed to higher exposures to triggering factors, as an infection or potentially after vaccination.

Second, our data showed that current classification of the six ADS subgroups share many clinical features, such as age of onset, duration of disease, cerebral spinal fluid protein, and cell counts, suggesting that these presentations may be a part of the spectrum or range of disease diversity. Additionally, based on the results presented in Table 3, it seems that distinguishing these groups from each other at the first presentation is impossible. Thus, our results suggest that current biological tests fail to identify a distinct subgroup of patients with overlap syndrome.

Third, TM/GBS overlap syndrome was the most common subgroup of GBS/ADS overlap syndrome in our study, in contrast to the very uncommon frequency of TM reported in 842 Western Australian cohort cases, which comprised only one study including both children and adults with ADS.⁵⁷ In addition, the frequency of TM in ADS children also has been reported to be uncommon in other studies.^{7,60} The higher frequency of TM found in the present study may simply reflect two factors: (1) there could be a discrepancy in defining TM; and (2) it is likely that the overlap between GBS and TM may share more common etiologies and genetic backgrounds. Taken

together with our observations, the latter factor is more likely to explain the higher frequency of GBS/TM; however, further investigations are necessary.

We found that the onset of GBS/ADS overlap syndrome occurred at the same time or that one disease (GBS or ADS) secondary to the other disease (ADS or GBS) occurred in the acute phase (<4 weeks). By definition, all patients with GBS had their nadir (maximal weakness) within 4 weeks, which was the reason that they were initially diagnosed with GBS. We further explored the timing of onset between GBS and ADS (Figure 1). The onset of this concurrent event mostly occurred at the very beginning of the course (<10 days in a total of 32 cases). Based on these findings, this study supported that GBS and ADS may form a continuous spectrum with variable PNS and CNS involvement rather than co-incidental occurrence.

The most frequent diseases in the ADS group, ON and NMO, caused a visual deficit. It is known that patients with GBS rarely present with a visual deficit; therefore, the clinical presentation of a visual deficit becomes a very specific sign/symptom in the diagnosis of GBS/ADS overlap syndrome among the five most common clinical presentations (e.g. paresis, sensory deficit, cranial nerve palsies, urinary disturbance, and visual deficit). During the early stage of the overlap syndrome, differentiating concurrent GBS and ADS from GBS alone can be very difficult. Based on our findings, one patient was diagnosed with GBS but with a later onset of a visual deficit. It is reasonable to suspect that this patient might have had GBS/ADS overlap syndrome. Additionally, clinical and other characteristics were not significantly different among patients except for sensory level (76%) in GBS/TM overlap syndrome.

Miller Fisher syndrome is considered to have very special lesion characteristics (ophthalmoplegia, ataxia, and areflexia), which are still subject to much debate regarding the peripheral or central nature of

Table 3 Biological and neuroimaging data

Variable	GBS/ ON = 13	GBS/ TM = 21	GBS/other CIS = 16	GBS/ NMO = 2	GBS/ MS = 7	GBS/ ADEM = 7	Total = 66
Cerebrospinal fluid							
Albuminocytological dissociation	9 (69%)	14/19 (74%)	8/16 (50%)	NR	7/7 (100%)	6/7 (86%)	45/63 (71%)
IgG index (n) >0.68	0/1	0/1	0/2	NR	NR	1/1 (100%)	1/5 (20%)
Positive oligoclonal bands (n)	3/8 (38%)	1/2 (50%)	1/9 (11%)	NR	2/3 (67%)	0/2	7/24 (29%)
VEP abnormal	10/10 (100%)	0/2	4/6 (67%)	1/1 (100%)	2/3 (67%)	0/3	3/10 (30%)
Antiganglioside antibodies	3/4 (75%)	2/4 (50%)	1/7 (14%)	NR	0/2	1/1 (100%)	7/18 (39%)
Brain MRI							
Supratentorial	0/10	0/7	5/14 (36%)	0/2	2/3 (67%)	5/6 (83%)	12/42 (29%)
Subtentorial	0/10	0/7	7/14 (50%)	0/2	0/3	0/6	7/42 (17%)
Both	0/10	0/7	2/14 (14%)	0/2	1/3 (33%)	1/6 (17%)	4/42 (10%)
Spinal cord MRI	0/3	17/19 (90%)	2/3 (67%)	2/2 (100%)	0/2	1/2 (50%)	22/31 (71%)

GBS: Guillain-Barré syndrome; ADS: acquired demyelinating syndromes; ON: optic neuritis; CIS: clinically isolated syndromes; TM: transverse myelitis; NMO: neuromyelitis optica; MS: multiple sclerosis; ADEM: acute disseminated encephalomyelitis; NR: not recorded; VEP: Visual evoked potential.

Data are expressed as mean (SD), median (range), or percent.

these syndromes, and such debate continues in the literature. Therefore, MFS/ADS overlap syndrome is an ideal model for studying mechanisms with underlying PNS and CNS involvement. In this study, we mainly identified the difference between non-MFS/ADS overlap syndrome and MFS/ADS overlap syndrome. First, this study accounted for a 20% frequency of MFS in GBS. Considering that the incidence of MFS among GBS cases is small throughout the world, the incidence of MFS in this study appeared to be greater. In this study, four of 11 MFS patients were from East Asia, where a greater frequency exists than in western countries.^{63,64} Second, based on the interval time analysis (Figure 1), most MFS patients (91%) seem to have ADS 'at the same time' (within a very short period), in contrast to 38% in non-MFS groups ($P = 0.0041$). In other words, these special subgroups identified seemed not to have a long enough interval time from MFS to ADS (and vice versa) diagnosis in most recruited case reports. Therefore, the interval time between diagnosis of MFS and ADS (and vice versa) was significantly shorter in patients with MFS/ADS overlap syndrome than that in patients with non-MFS/ADS overlap syndrome. Taken together, our findings confirmed that these MFS patients developed a more homogenous and mixed clinical picture within ADS. Furthermore, we found that MFS/ADS (in fact MFS/ADS is equal to MFS/other CIS as defined in this study) patients showed a significant subtentorial predominance according to the brain MRI. Although we classified ADS into six categories arbitrarily, we suspected the features of MFS/other CIS may more reflect involvements of the brainstem (one of the nosologic position) in MFS itself, other than ADS in autoimmune peripheral neuropathies. The nosologic position of MFS is unclear, with

speculation being that the lesion site in MFS is in the peripheral nerves, brainstem, or cerebellum. The role of brainstem involvement in MFS was also observed in neurophysiological studies.⁶⁵ Nevertheless, it has previously been indicated that MFS can be referred to as the 'anti-GQ1b antibody syndrome'.⁶⁴ Within the CNS, an anti-GQ1b monoclonal antibody has been reported to weakly stain deep cerebellar nuclei, the gray matter in the brainstem.⁶⁶ Our finding thus supported this prior pathological observation.

Based on the clinical studies analyzed in this study, the sensory level was the best predictor of a poor outcome, while the visual deficit predicted a favorable outcome. The sensory level is the most common presentation in the TM group, but it occurs much less often in other subtypes of ADS, whereas visual deficit is found in all ADS subtypes except for TM. Therefore, it seems that the poor or favorable predictors reported here could more realistically reflect the true lesion load of one separate subtype of ADS in this study.

A primary strength of this study was that we summarized and reviewed data from the largest series of patients with overlap syndrome, which provide good insight of the spectrum of this overlap syndrome, whereas prospective cohorts are virtually impossible to perform on rare diseases. Furthermore, we believe that the application of strict diagnostic criteria of ADS according to the consensus definitions gave the best possible correction for the estimated frequency, which should be more comparable with future research.⁶ However, our study also had some limitations. First, this study was not allocated a control with classical GBS or ADS. The ideal experimental design was not feasible because of the inherent features of the included case reports. Therefore, our design was the second best alternative.

Table 4 Comparison between MFS/ADS and non-MFS/ADS overlap syndrome

	MFS/ADS	Non-MFS/ADS	F*/OR	P
N (%)	11 (18%)	55 (82%)		
Sex (M:F)	2:9	31:24	5.813	0.021
Mean age (SD), years	45.5 (19.8)	27.9 (21.0)	6.552†	0.013
Simultaneous initial diagnosis	10/11 (91%)	21/55 (38%)	8.224	0.004
Interval between primary to secondary diagnosis				
≤4 weeks	11/11 (100%)	46/55 (84%)	1.196	0.149
Anti-GQ1b antibodies	2/3 (66.7%)	2/10 (20%)	8.000	0.203
Brain MRI				
Supratentorial	0	12/31 (39%)	6.475	0.011
Subtentorial	6‡ (86%)	1/31 (3%)	20.659	0.000
Both	1§ (14.3%)	3/31 (10%)	1.327	0.249
Spinal cord MRI involvement	0/1	22/30 (73%)	7.339	0.007
Poor outcome	0/10	17/54 (32%)	6.050	0.014

MFS: Miller Fisher syndrome; ADS: acquired demyelinating syndromes; OR: Odds Ratio.

*One-way analysis of variance.

†Data are expressed as mean (SD), median (range), or percent.

‡Involving lesions include lateral lower medulla on both sides (1 case), the posterolateral of the pons (2 cases), midbrain tegmentum (1 case), left paramedian mid-brain extending to the thalamus (1 case), and left mesencephalon (1 case).

§Involving lesions include bilateral cerebral white matter, posterior brainstem, and left cerebellar (1 case).

Second, the exclusion of cases with missing data will impact a change in the denominator. Third, we used the Hughes scale as an outcome measure; readers may argue that this scale will more reflect GBS in this overlap syndrome than ADS. However, currently, due to the lack of a consensus measure for ADS, the GBS disability score seems to be a suitable alternative measure, especially in this study which was performed retrospectively based on case reports. Furthermore, given the low frequency of GBS/ADS overlap syndrome, we might have missed a stable estimation and the data available may lack the power to detect clinically meaningful factors. Last but not least, journals are more prone to accept for publication unusual cases that appear to be intriguing and unexpected. This certainly causes bias in frequency estimation in regard to the associated clinical and demographic findings. Nevertheless, we emphasized the importance of detection and management of GBS/ADS overlap syndrome, which could improve the outcome.

Conclusions

The ADS in GBS are diverse, CNS demyelinating may occur at any time, but early in the GBS course (and vice versa). Miller Fisher syndrome/ADS overlap syndromes is more common. The prognosis is generally good, but patients with sensory level deficit are likely to have a poor prognosis. The other CIS in MFS may be one of the nosologic position, rather than 'ADS in autoimmune peripheral neuropathies.' We emphasize both the clinical and prognostic findings of GBS/ADS overlap syndrome, which will help us to further understand the nosological relationships and to improve the outcome of the overlap syndrome.

Disclaimer Statements

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Ethics approval The treatment for the patient was carried out according to the protocol approved by Institutional Ethical Committee of Affiliated Nationalities Hospital, Guangxi Medical University.

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